

CHAPTER TWELVE

Requirements from Groups

Negotiation builds a team as well as a set of requirements.

Barry Boehm

Requirement Elements	Stakeholders	Goals	Context, Interfaces, Scope	Scenarios	Qualities and Constraints	Rationale and Assumptions	Definitions	Measurements	Priorities
Discovery Contexts									
Introduction									
From Individuals									
From Groups									
From Things									
Trade-Offs									
Putting it all Together									

Answering the questions:

- What do you collectively want?
- What conflicts are there, and how can you resolve them?
- How can groups divided by distance, culture and organisational barriers work together?
 - ... so you get to hear how people's points of view fit together
 - ... so you discover potential conflicts early enough to be able to do something about them
 - ... so the team comes together and agrees the requirements.

12.1 Summary

Well-run **groups work as systems**: together, **they can do more than all the individuals** within them, benefiting from interactions. Group understanding, therefore, needs to be discovered from the whole group.

Workshops bring a group of people together, with a common aim of finding a solution to a shared problem on which they have different visions.

Requirements workshops bring together a group of people with a common requirements discovery aim, such as understanding a process, agreeing a set of goals or working out a rationale.

Obstacles to group work include geographical, cultural and language barriers.

Group media and groupware – technologies that facilitate group interaction – may help people to discover requirements despite obstacles, but they need to be carefully managed.

Group work demands both technical and social skill, especially for collaborative requirements activities such as workshops. Detailed suggestions are made in this chapter for planning and running a workshop, and for engaging workshop participants.

12.2 The Goal of Group Work

The purpose of working on requirements through groups is to get **enough knowledge and energy together** to uncover requirements quickly and efficiently, especially where no individuals can do that on their own.

12.2.1 Unique Capabilities

To some extent, group work overlaps with working with individuals (see Chapter 11) in its capabilities for discovering requirements. But groups can assemble information from many sources, fit it together, hear many points of

view, refine the collective understanding and reach agreement in a way that simply is not available any other way.

The traditional, and still the principal, mechanism for groupwork is for people to work together: a team sitting side by side in an office; an ad hoc group assembled around a table in a workshop or conference.

12.2.2 Obstacles

Increasingly, teams are split across two or many sites, and sometimes across boundaries of language, organisations, cultures, nations and time zones. These are serious impediments. Face-to-face meetings and co-location of teams may still be the best way of overcoming such obstacles. But these are costly, so projects have tried a wide range of approaches for group work mediated by tools, generally involving software.

The need for confidentiality (an aspect of security) is potentially a serious obstacle to group work and communication between sites, when requirements must be protected against disclosure to negative stakeholders such as competitors. Security rules may forbid electronic communication across the fence of a military site. Some negative scenarios (see Chapter 5) may relate to people inside the project or organisation, in which case they cannot be discussed openly in the group. Technologies such as encryption can help to overcome some of these obstacles.

12.2.3 Mediating Group Work (on one site or many)

Information and communications technologies – computers and networks – offer many possible ways to mediate group work, for example by sharing models or requirements, by virtual conferencing, by creating a project website or by Wiki.

Many of these ‘groupware’ tools are useful to every project, whether working on one site or on many. But how far they can be a substitute for face-to-face working depends on the quality of the ‘togetherness’ that they can provide.

This quality is often not very good, for reasons that are not just technological. For instance, working between time-zones that are five or more hours apart is simply difficult, as the number of hours available for meetings is limited. We suspect that a mixture of mediated and face-to-face group work is necessary for most projects.

In the following sections, we look in turn at:

- **workshops** – specialised requirements discovery meetings;
- **group media and groupware** – tools (software or not) that promise to mediate requirements discovery.

12.3 Workshops

A workshop is a specialised meeting, structured to bring exactly the right people together to solve a problem using a planned sequence of activities. Those people may be any stakeholders in the project, members of the development team or, if need be, external experts to assist with specific tasks.

12.3.1 Define Workshop Mission

Treat a workshop like a project: it has a mission, requirements and an implementation phase. A good workshop sets out to achieve a definite aim, and is planned and executed to achieve it.

A suitable mission for a requirements workshop can be almost anything on a project that needs the skilled involvement of stakeholders and requirements specialists, e.g.:

- to define the goals for the XYZ product;
- to create scenarios for the XYZ product;
- to draft the performance requirements for the XYZ product.

Workshops are probably the primary way to get decisions made and agreed by groups such as project teams.

Workshop activities produce requirement elements such as goals and scenarios, as described in Part I of this book. Therefore, to plan your workshop's activities, you should refer to the appropriate chapter of Part I, as well as to the element-independent advice in this chapter.

Workshops are not 'Meetings' (let alone Presentations)

Never think of a requirements workshop as just a meeting. Meetings may be called ad hoc, with no planning or structure, to address an immediate issue. They may involve presentations made by specialists, or by managers. Some business experts think ad hoc meetings and slide shows are always undesirable. But whether or not such a top-down approach works for meetings, it is quite unacceptable for requirements workshops.

Instead, we have two basic 'equations' in mind:

$$\begin{aligned}\text{Workshop} &= \text{Meeting} + \text{Purpose} + \text{Planned Activities} \\ \text{Specific Workshop} &= \text{Workshop} + \text{Element-specific Activities}\end{aligned}$$

12.3.2 Workshop Planning

Workshops seem to work best when they are planned in detail but allowed to run quite freely. The use of time is critical, as an N-person workshop costs at least N salary-days to run, and people can quickly become bored.

Workshops as Theatre

A workshop is a kind of performance. Many people will attend, so each minute of workshop time costs many minutes of salaried time. Each participant wants to benefit from the time invested. Each participant will make a judgement on whether your workshop – and you – were worth their time.

It is therefore better to rehearse your plan in ordinary 'slow' time, and discover your mistakes in private, rather than to make costly mistakes in workshop time.

Workshops are like theatre, too, in that people want to enjoy the time, as well as to get results. Getting stakeholders' 'buy-in' is often critical to success. An element of showmanship is necessary. That means planning, rehearsal, and playing the role.



Direct use of improvisational theatre ('improv') techniques is also possible (scenarios, roleplays etc); or, improv can be used as a training aid for better team-based innovation performances (see Martin Mahaux's guest box). Simple techniques, such as acting a scenario around a flipchart model of a user interface, can be very effective.

- Budget the time carefully, and plan the workshop activities both for managing overrun (can some activities be dropped or shortened?) and underrun (should you finish early, or take along some 'spare' activities?). Think through and discuss the scenarios.

- Work out who to invite: all the people you need for the planned activities, and no passengers. Then, tell those people about the workshop and agree the date. This conveys the essential message that you care that the invitees attend.
- The planning of refreshments and meals is critical, as the energy of a workshop is swiftly dissipated if people melt away to do their email every time there is a break, and only return much later. Therefore, coffee, water, snacks, etc should be brought to the workshop room at planned times.
- The suggestions here are all based on experience. If you follow them carefully, define your goals for the workshop, plan what you want to happen and rehearse thoroughly, your team should be able to run a successful workshop. However, if running your own workshop seems too daunting, hire an experienced requirements workshop facilitator.

Guest Box: Mindset And Skills For The Requirements Workshop

by Martin Mahaux, inno.com

A successful requirements workshop requires careful preparation, but also a specific mindset and specific skills. Your chance of running a successful requirements workshop depends heavily on the ability of the facilitator and the participants to **innovate as a team**.

First, we need a mindset **attuned to a common objective**. The stakeholders share a common objective, but have different visions of the problem and very different sub-objectives. The common goal is to make the project a success, but the business sub-objective may be 'more features' and the security guys will strive for 'more resistant to attacks'. Of course, nobody should forget their own sub-objectives, but the *credo* of a workshop participant should be:

'My point of view is one amongst others,
and the most important is the final common objective.'

You are very likely not to achieve anything if you can't achieve this.

At the level of your project, you can take various efficient actions to foster such positive culture. Think about it when you organise team building events; show the example when you organise team meetings; make sure people meet face to face sometimes; hook posters in the room to promote the feeling of belonging to the team; try not to segregate project roles too much. Advice on how to care about the social factors on

a project can be found in Beyer and Holtzblatt (1998) [2], Cockburn (2006) [3] and Boehm *et al* [5].

At the beginning of the workshop, ensure the facilitator recalls the desired focus on the common objective. Also, make use of techniques or methods for fostering team collaboration. Some easily applicable and efficient techniques can be found in Ellen Gottesdiener's (2002) book [1].

Once the good mindset is in place, participants need the tools to make the magic happen: the '**innovative team playing**' skills. In the toolbox are communication skills (in both directions: make yourself clearly understand, listen actively), negotiation skills, flexibility, creativity and interpersonal skills.

You know books or classical training programs won't help in providing these tools, as skills can only be learned by experience. Well, there is an unexpected place where innovative team playing experience is a discipline: **improvisational theatre** or '**improv**'.

The principle of improv is simple: a topic is chosen and, instantly, actors begin to play together a new piece around that topic. Improv sessions are like requirements workshops: the different actors enter the play with different visions and ideas about the story they will build together, with the common goal of making it a great moment for the audience, here and now, together. Improv is nothing but pure innovative team playing experience. It is also extremely enjoyable, and the discipline has developed proven training techniques to teach it step by step. So, why not use it when training requirements people? I tried it in a tailor made training program, and can't give you better advice than to try it yourself! People will learn things about themselves and about others, and will be ready to work together to build innovative requirements.

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12.3.3 Workshop Rehearsal

A literal dress rehearsal is not usually necessary for requirement workshops. You are more likely to benefit from mini-rehearsals, such as:

- thinking through what you will say at each stage, and perhaps practising in front of a mirror if you are unused to speaking in public;
- trying out any group exercises such as brainstorming sessions. A rehearsal can reveal answers to concerns such as: is there, in fact, anything much to discover? How long will be needed? What resources will people need?;

- anticipating likely questions and challenges, and preparing materials to handle them;
- checking that you can be seen and heard from the back of the room.

12.3.4 Workshop Setup

The careful preparation of materials and setup is a critical part of running a successful scenario workshop.

Tested Resources

The actual resources you will use must be instantly available, and tested on the spot so you know they work.

- If you are using electronic equipment like computers, projectors or electronic whiteboards, test it, and ensure you have spare equipment to hand.
- Workshops do not have to be run with any electronic equipment; they can work better without, as the human contact is then more natural. Writing and drawing on flipcharts can be fast and effective.
- Check the supply of paper and pens; flipchart pens usually contain permanent ink, which is a small disaster on whiteboards.
- If you mean to draw on whiteboards (necessary if you need to make large drawings), take a digital camera (with spare batteries and plenty of free memory) to record the results.
- If participants are to be provided with reading materials such as background notes, make sure these are printed and bound in good time. Ensure they arrive at the venue safely: it is no good if they get lost in the 'goods inwards' department on the day of the workshop.
- If you are using a venue like a hotel, and you depend on them to photocopy materials for you quickly during the workshop, test them out first to ensure they really can respond reliably. Venues sometimes advertise photocopy and fax facilities when all they have is a slow multipurpose machine at reception, which reception staff are too busy to operate.

Room Setup

- The workshop room must be set up in advance.
- The traditional room layouts of a circle or U-shape work well, as they imply equal status for all participants.

- If possible, set the room up the day before; failing that, at least look at the room, work out how to set it up and arrive early to shift tables, etc.
- Aim to give everyone the feeling that you know exactly what you are doing, that the workshop is well planned and that their participation is important.

Fix Problems in the Room

Ensure that you discover and fix any problems in the room. Common ones are listed in Table 12.1.

Setting 'House Rules'

- There is usually one person in a group who likes to talk continually, so plan what to do to prevent that. For example, state the 'house rules' at the beginning, emphasising that time is limited, and you reserve the right to park issues for discussion after the workshop.
- Minimise other disturbances:
 - ask people to switch off their mobile phones, etc;
 - stress that you require everyone's full attention, as their silence will be taken as complete agreement to all decisions taken during the workshop.
- Set roles for people:
 - **Time-keeper** to announce 'coffee time', etc. This frees the facilitator to focus on discovering requirements but, more importantly, it says that the workshop is a communal effort, not a performance by one or two people.
 - **Rat-hole watcher** (as suggested by Beyer and Holtzblatt (1998) [2]) to announce 'this is a rat hole' when the discussion is wandering way off the subject (down an infinitely-branched and well-worn but irrelevant hole). This implicitly gives everyone, not only the official hole-watcher, permission to look out for rat holes.
 - **Dictionary owner, issues list owner, data modeller**, etc. These people can share the work of recording and organising workshop outputs. But again, the real purpose of doing this is deeper. It is to alert people to the fact that arguments over terms, or anxieties over project issues, can be raised consciously as things to be addressed (rather than traps to fall into).

Table 12.1: Solutions for problems in workshop rooms.

Problem in the room	Solution
Poor lighting (e.g. inadequate blackout; lamps pointing straight at the projection screen).	• Visit the room. • Try out the lighting. Make sure you can read what is projected from the back and sides of the room. • Ensure that all the issues are addressed before the workshop.
Inaccessible power supplies.	• Bring extension cables and gaffer tape. • Ensure the cables cannot cause people (including you) to trip. Tape them down well.
Broken electronic equipment such as projectors; Incompatible equipment; Video and network cables with bent connector pins; Network sockets switched off, etc.	• Test everything as if you were holding the workshop. • Assume Murphy's Law applies ('If anything can go wrong, it will').
Excessive amounts of furniture.	• Ensure the room only has what you need, with the maximum of space to move around. Stacked chairs and tables give an immediate impression of clutter and chaos – exactly what you don't want in a workshop.
Poor ventilation, heating or cooling.	• Find out how to open windows and work the air conditioning in advance. • Ensure there is water for everyone to drink. • If the air conditioning is making participants uncomfortable, turn it off. • Remember that seated participants can become much colder than active facilitators, especially if they are sitting right under an air conditioning duct.
Inadequate wall space, or forbidden to pin/stick flipcharts to the walls	• Find a room with blank walls, glass partitions or other surfaces you can stick things on to. • Technological fixes include: hanging backing-paper (e.g. newsprint) to stick things on to; electrostatically-charged folio flipcharts that stick reversibly to any surface; temporary screens or pinboards.

Guest Box: How to Engage Participants in a Workshop

by Joy Beatty, Seilevel

Workshop sessions should be kept as short as possible, ideally one to two hours. However, some projects will require longer sessions, sometimes for multiple days in a row. Day-long workshops are at high risk for being tiring, boring, and uninspiring. To combat this, use activities that will keep participants mentally alert, while encouraging them to actively think about the topics and share ideas.

The following suggestions are specific examples to help engage workshop participants.

Move Around

Participants will stay more attentive in workshops if there is active movement in the room, and it is best if they are part of the movement. They can take turns writing notes on the wall, to keep them alert and give them ownership over the content. The official facilitator and scribe will still be responsible for making sure the substitutes are doing a quality job, with subtle guidance as needed. The person at the wall should *not* be the lead expert on the specific topic being discussed.

When the energy dropped in a process flow workshop for engineers, the scribe at the wall tossed the whiteboard marker to another engineer to take over. Trading-off the scribe role continued throughout the session. The participants enjoyed the opportunity to participate in some way other than sitting down.

Play Games

Games encourage cooperation and friendly competition, which help engage participants and inspire insights. Games should be used to produce valuable content without slowing the workshop down significantly. The game rules must be simple. It helps to have small prizes for participants who offer particularly useful insights or active participation, so that the emphasis is not just on winning.

A game was used to identify features using affinity diagrams, with a twist that required silence (based on Thiagi's *Silent Movies*¹). The workshop attendees had blank sticky notes, on which they listed individual features that came to mind. Then, without talking, in small groups they categorised the collective set of features on a piece of paper on the wall.

¹Thiagarajan, S. (2003) *Design Your Own Games and Activities: Thiagi's Templates for Performance Improvement*, San Francisco, CA: Pfeiffer

If the group members disagreed on features or categories, they had to argue *silently* to come to resolution. In the end, the facilitator brought all the papers together, leading verbal discussions focused on the quickly identified differences.

Use Toys

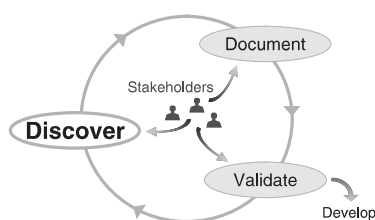
Placing toys on the workshop tables supports a fun and relaxing environment, while alleviating fidgeting and fostering creativity. It is important that the toys are kept simple, so as not to be too distracting. It is best not to pick mentally challenging puzzles that require concentrated attention to play.

For example, Play-Doh® (a modelling paste) was given to a group of executives who were brainstorming product innovation ideas. They were asked to model with Play-Doh what ‘product innovation’ meant to them each personally. The models they created included a dollar sign, a box with a ball that did not fit in the box, Yin and Yang, and a window to the world. The activity revived the creative minds in these executives after a number of hours of concentrated thinking.

As a final suggestion, activity selection should be based on the type of attendee, the goals, the context, and the schedule of the workshop, so as to create an effective and engaging experience.

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Workshop Activities



Workshops can provide the context for discovering (or starting to discover) many of the types of requirement described in Part I of this book.

For instance, your project might need to use one or more of these activities:

- **Stakeholder onion/influences model (Chapter 2):**
 - Draw a diagram of the stakeholders involved, and the influences they have on each other.

- **Brainstorming and clustering goals (Chapter 3):**
 - Ask participants to write their goals on sticky notes in, say, 5-10 minutes.
 - Ask participants to stick their notes on the wall.
 - As a group, cluster the goals into groups on the wall, and give each group a title.
 - Have everyone sit down, and take feedback on the activity.
 - Give everyone a second chance to write down any missed goals.
 - Cluster those goals also.

Emerging Issues

Do not be disappointed if a workshop output, such as a cluster of sticky notes, is not what you would call a goal. If a workshop is more concerned with issues like lack of training, or the difficulty of working across multiple sites, then that is what they immediately want to deal with. Of course, the *project* goal 'provide training' is the flip side of the issue. There is no reason why all the goals should be for the *product*.

Once you have respected and dealt with people's concerns, goals will emerge without effort.

- **Goal model (Chapter 3):**
 - Brainstorm the goals of different stakeholders.
 - Draw a goal model to identify synergies and conflicts.
- **Threats, risks, obstacles, and countermeasures (Section 3.2.6, The Negative Side, in Chapter 3):**
 - Identify negative aspects by brainstorming, war-gaming ('I could defeat that by doing XYZ'), walking through scenarios or scanning a template of known risks.
 - Identify countermeasures to each negative item.
- **Context model/rich picture (Chapter 4):**
 - Draw whichever kind of context diagram is most appropriate, showing what lies just outside (and if need be, what lies just inside) the boundaries of the system. Explore the possibilities by moving the boundary to include more or fewer items.

- **Scenarios (Chapter 5):**

- List scenarios on flipcharts or whiteboards, and be ready to correct them as you find out more. There are several useful workshop techniques for scenarios; scenarios and workshops go well together, forming a quick and powerful means of discovering and agreeing requirements with stakeholders.
- For detailed suggestions for scenario workshop activities, see Chapter 6.
- Ellen Gottesdiener's (2002) *Requirements by Collaboration* [1] makes many practical suggestions specifically for scenario workshops.

- **Qualities and constraints (Chapter 6):**

- Workshop with business users, scanning a template of typical NFRs.
- Work in specialist areas such as security or safety; for instance, a safety team could run its own workshop.
- Run a show-stopping constraints workshop, e.g. to bring together people from different areas to identify the top risks to the project.
- If not already done, run a goals workshop to agree the key qualities of a product.

- **Rationale and assumptions (Chapter 7):**

- List the major assumptions that participants are making in their scenarios, or in other documents.
- Create rationale diagrams to explain the reasoning behind any critical decisions facing the project, or to capture any arguments that arise, in a neutral way.

- **Terms/definitions (Chapter 8):**

- Once you have already held a scenarios or other workshop, you may find you need to run a workshop activity (probably not often a whole workshop) to resolve terminology issues.
- Make a list of project dictionary terms, and get participants to define each term. If there are competing definitions of some terms, invent new terms to go with each definition. (Also watch out for synonyms, etc.) Disagreements on terms can be very revealing; resolving them can quickly lead to progress.

- **Priorities (Chapter 10):**

- Debate, vote, sort cards, cluster sticky notes, etc, to obtain agreement on the importance of the requirements to stakeholders.

- **Trade-offs (Chapter 14):**

- Brainstorm possible design options.

- Score the design options, once these have been worked out in sufficient detail, on the criteria (based on requirements).

Tips for Workshops

- Plan carefully.
- Test anything that seems risky or uncertain in advance of the workshop.
- Use techniques that you know will work for you – live workshops are not the right place to experiment.
- Visit the venue and do a ‘dry run’ as if you were running the workshop.
- Talk through the schedule with the venue manager – make sure you agree what is going to happen, and when.

Combination Workshops

The most basic advice for workshops is ‘do one thing at a time’.

But it can be right to work through several activities, involving more than one requirement element (from Part I of this book) for a single workshop.

For example, one successful workshop went like this:

1. **‘Theatre’** sketch setting the scene (a few minutes).
2. **Scenarios:** recorded from each of three areas of work involving apparently the same general kind of problem, but which were rightly suspected of having strongly differing equipment needs (most of a day).
3. **Brainstorming of goals** derived from all participants’ reflections on the scenarios they had just witnessed (half an hour).

Complex Topics

By their nature, complex topics are all different. Often, you can split a complex topic into a series of simpler questions, such as:

- What are the stakeholders’ goals for this? (Chapter 3)
- What conflicts have arisen? (Chapter 3)
- How do we see this working? (Scenarios, Chapter 5)
- What are the arguments on each side? (Chapter 7)
- Can we imagine different resolutions to the problem?
- What are our priorities here? (Chapter 10)

Such questions will often match a chapter in Part I of this book; and they form likely candidates for separate workshops – or at least, major activities within a workshop.

Perhaps the most common complex question that faces projects is that of selecting an approach that best meets a set of goals, i.e. trade-offs (Chapter 14). That is a matter of both requirements and design, so a resolution will usually involve different teams, often from different organisations. Trade-offs are good subjects for workshops, but these demand very thorough preparation, including the design of different options (not just the preparation of the workshop as a meeting).

Large Groups

Large projects usually mean complex problems and large teams: a poor mixture for a workshop. If possible, meet people individually first, and hear their points of view.

Age-old wisdom is that effective teams and meetings are quite small. A general, for instance, commands a huge army, but his planning meetings involve only a handful of people, who all know each other well. Parliaments consist of hundreds of delegates, but government is by a cabinet of ministers who can all sit around a table together. You can't have a shared decision-making workshop that fills a lecture theatre.

It is therefore critical to find a way of bringing group size down to an effective number. Even 20 is becoming large for a workshop. After all, with 20 people in a room, each person can speak for three minutes per hour on average; and many people will not speak at all. Facilitating a workshop of 30 people is possible, but only just. As you add more participants, the meeting fatally changes its character from a participative event to a lecture or performance.

We have attended, and occasionally been forced to run, supersize workshops of various kinds. The organisers try various tricks to give people a chance to feel as if they are getting involved: splitting into breakout groups, each with a topic to examine; appointing a spokesperson to report on the group's findings; preparing one or two slides of summary; presenting the slides in a rushed, two-minute slot. The overall result is a mass of detailed organisation on the day; a lot of 'Now we will call you back into the main room in 10 minutes', plenty of shuffling about and rather inconclusive results.

So, workshops must be small. Small teams should run separate workshops. Findings can be reported to other teams. Team leaders can get together. Interfaces between two teams can be agreed locally. These technical activities should be quite separate from any desire to get everyone together for other purposes.

Not to be Attempted

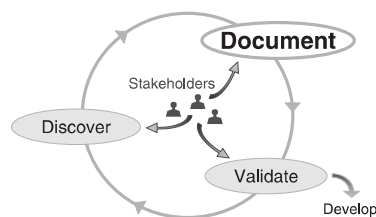
It is hard to legislate on what not to do in a workshop; some of the best workshops achieve success with agendas that look impossibly ambitious.

There is value, too, in a degree of hustle; time flies in the best workshops.

But, in general, do not try to do too much. Having a single goal, with several related activities, is more likely to work than a hybrid mishmash of a workshop that attempts to do everything.

We have emphasised workshop planning. But workshops can succeed under pressure, and may thrive on it; planning can be done quickly, as long as good thinking goes into it.

12.3.5 Workshop Recording



The task of recording a workshop's findings is important. An accurate record, shared promptly with all participants, builds trust that the process is fair.

A full-day workshop (for instance) can generate a large volume of findings. You have two conflicting goals:

- to record every finding;
- to allow the workshop to proceed smoothly (at full speed).

Drawing by Hand

You can draw diagrams by hand and record them with a digital camera (if allowed), on a self-printing whiteboard or simply by keeping flipchart sheets. This is probably the best approach for a workshop. Drawing with pens is often faster and less fiddly than using software. It is also more reliable. But the main advantage is that it supports human communication better. For example, participants can see that you draw and label their bubble when they call out a goal. They understand that they are being heard. This feels quite different from presenting people with a clever model that you dreamed up. You can quickly correct mistakes on a whiteboard, too.

Your solution will vary, but it will involve:

- enough effort to document results as they appear (e.g. **scribe** with computer, **facilitator** with flipchart pen, etc);
- appropriate choice of recording tools and techniques (flipcharts, whiteboards, writing materials for participants, camera, voice recorder, etc);
- a suitable set of activities for participants, some of which will involve them writing things down, drawing sketches, acting scenes, etc.

Flipcharts have the great advantage that they can be put up on the walls of the workshop room, offering immediate feedback that the workshop is being productive, and participants are being heard.

Guest Box: A Three-Person Facilitation Team

by Dr Frank Houdek, Daimler

On a recent workshop, I observed a novel workshop approach with a larger than usual team of facilitators.

The **moderator** documented everything on flipcharts. The flipcharts were put up on the walls so they were visible to participants all the time.

A **scribe** from the client organisation recorded the findings on a laptop.

An **observer** (from the same consultancy company as the moderator) sat beside the scribe and helped her. His major role was to act as quality control. If he got the feeling that the moderator had got something wrong from the participants' statements, he joined in discreetly. He asked questions like: 'Maybe I got it wrong, but ... ', leaving the moderator in the 'active position'.

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Recording a multi-activity workshop, especially if there are parallel streams, is complex. Plan and rehearse how you will record and collate all the findings. For example, you could decide to label each flipchart sheet with the date and activity that it records.

Recording with a Digital Camera

A digital camera is a huge help in recording workshop findings.

Common problems include:

- glare from the reflection of the flash on whiteboards, affecting readability;

- poor quality images (odd colour, low contrast, distortion, fuzziness);
- inconveniently large image files.

A few tips will help you get much better results:

- Stand slightly to one side to avoid reflections.
- Shoot a test photograph; if the flash causes problems, turn it off.
- Process images to give them a standard, good quality appearance.

Image processing need not be complicated. Work out a simple approach that does the job for you. You will need a 'paint shop' style software package (one is often provided free with digital cameras). A basic sequence is:

1. **Perspective correction** (to make whiteboards appear as undistorted rectangles). Typically, the tool shows a rectangle in front of your image. You stretch its corners to match the corners of your trapezium-shaped whiteboard, and double click to apply the correction.
2. **Colour balance** (use the 'fluorescent bulb' or 4200 Kelvin setting for room lighting).
3. **Contrast enhancement**.
4. **Unsharp mask**.
5. **Crop** (to remove blank areas around the edges).
6. **Save as** a JPEG image with medium compression.
7. If the file size is still very large, save a copy of the image, and **resize** the copy to (say) 30% of the original. Sharpen it and check it is still legible.

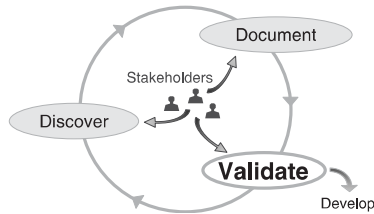
Many of these steps may be available on your package in a command named something like '**One Step Photo Fix**'. Better, your package will probably be able to record a sequence of commands as a script that you can re-run with a single command.

Make an index or overview document to help people find their way around the JPEG images. This can be a Word document, or a simple web page with named links to the images.

Document files may become very large if you drag images into them. Word files stay much smaller if you use Word's **Insert, Picture, From File . . .** command. It helps to keep groups happy if you send them small files.

Finally, there is a trade-off between sharing images quickly and taking a little time to translate the images into proper models. Don't spend hours on image processing.

12.3.6 Validating Workshop Findings



A good workshop should leave you with the feeling that you understand what the stakeholders want, along with a detailed (if handwritten) record of the workshop's findings. This is a human thing, a warm, shared assurance based on the trust developed in the workshop (see the guest box on 'Building and Sustaining Trust', by Ellen Gottesdiener).

Therefore, validation should mainly consist of checking that you have not lost anything when tidying up the findings afterwards. This is most simply done by asking the participants to check your workshop report.

Guest Box: Building and Sustaining Trust

by Ellen Gottesdiener, EBG Consulting

A fundamental ingredient of successful requirements workshops is building and sustaining trust. This may seem 'touchy-feely' and vague, yet there are mechanics for trust building. I find them consistently useful in the workshops I have planned, designed and facilitated, and I use them in every workshop.

Some of these techniques are necessary to employ before, during, or after the workshop event. I associate them with three types of trust: contractual, communication and competency.

Building trust before the workshop

Form a planning team composed of a balance of perspectives (business and technical) that will assist the facilitator in planning and design. Craft a clear purpose that resonates with all workshop stakeholders. From there, define the workshop products or outcomes. Most of your workshop products will be requirements in some form, but your outputs will also include decisions (e.g. what is in and out of scope, what the business rules are for each set of scenarios, the priority of requirements, what stories to deliver in a given iteration, and so on).

Let's face it: requirements development is a series of ever-unfolding decisions. Prior to the workshop itself, communicate to all stakeholders what the decisions need to be, what decision rules the group will use,

and the decision-making processes the facilitator will employ to help the group reach closure.

Defining your purpose and products (including decisions) builds **contractual trust** – that is, trust based on clear expectations, mutual interdependency and joint accountability. Contractual trust is built by delineating workshop roles and responsibilities early on (e.g. the facilitator, recorder, participants, sponsor, observers and support cast).

Building **competency** is another ingredient for trust: the ability of people to rely on each other, seek input, and obtain feedback from each other. This means getting the right participants, including a mix of subject matter experts, customers, and technical staff – the people who know the problem domain; those who will use the product; and those who will build, supply, support and maintain the solution.

Another type of trust, **communication trust**, involves a group's ability to enjoy open, honest, and frequent communications (truth telling, with good purpose). Set the stage for communications trust by actively seeking out hidden agendas, so you can prepare appropriate activities or obtain permission to surface difficult issues or 'undiscussable' topics during the workshop.

Building and sustaining trust in the workshop itself

Start by having the sponsor present and briefly 'kicking off' the workshop. She or he should state why the workshop is important, expectations of the group's work, and how they will support those outcomes. This adds to contractual trust.

After the kick off, and as part of setting the stage for getting down to work, workshop participants should establish working agreements or ground rules for how they will conduct themselves during the session to building contractual and communications trust. Include decision-making rules, which in trustful groups is transparent and explicit.

One of the facilitator's greatest challenges is facilitating in the face of conflict. Welcome conflict! It's a normal part of group dynamics, and its absence may indicate suppressed issues likely to surface during the workshop. Conflict is also an indicator of energy to be harnessed for productive ends. Workshop participants build communications trust when they directly and honestly discuss conflict. By using agreed methods for making decisions to achieve closure, and openly addressing misunderstanding and disagreement, the facilitator helps the group manage conflict and boost communications trust.

Effective requirements workshops rely on participants' interdependent knowledge and skills. Playing with multiple interwoven analysis models (e.g. scenarios or stories, business rules, data model, actors and so on)

builds the group's competency with the problem domain, while concurrently maintaining effective communication. Requirements workshops require continual learning and feedback. When participants 'play' with multiple requirements models, they push the edges of their knowledge and skills, and learn from each other. Multi-modeling verifies requirements. At the same time, participants test each others' inferences and assumptions about those requirements. This builds both communications and competency trust.

Sustaining trust post-workshop

I recommend conducting a sponsor or key stakeholder 'show and tell' at the end of the workshop or series of workshops to increase communication trust up and down the organisation. These 'show and tell' events (which can be conducted a day or two after a workshop, if necessary) increases management's competency with the requirements. Since management folks often don't participate in detailed requirements workshops, yet need to understand the impact, risks and trade-offs associated with them, hearing outcomes in a 'show and tell' increases their knowledge so they can better support the project.

Be sure to share the workshop documentation with all participants as well as the larger stakeholder community soon after the session, thereby sustaining communication trust.

Your final workshop activity should be a workshop retrospective (i.e. debrief, self-reflection, lessons learned) that takes a hard look at how the workshop went, what worked well, what could be improved, and actions for improvement. The retrospective explores both product (workshop deliverables) and process (effectiveness of group interactions and workshop planning and communications). Retrospectives build on all three types of trust – contractual, communication and competency.

I also recommend conducting a post-workshop retrospective two to four weeks after the workshop or series of workshops. Now that some time has gone by, do we still think that the workshop was a good use of our time? What did we do really well? What deliverables from the workshop are we using now to build, test, and deploy the product? Did we get the right ones during the workshop? In retrospect, do we have the right people?

Reflecting on the utility of the workshops helps you build overall organisational competency with requirements workshops. After all, we want our workshops to be positive and productive, and truly achieve an experience in which, as Aristotle remarked, 'the whole is more than the sum of its parts'.

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The output from a workshop often involves various models from Part I of this book (e.g. goal models, scenarios, rationale models). You should check those using the techniques listed in the relevant chapters of Part I.

12.4 Group Media

Documentation is the worst form of communication.
Alistair Cockburn

Group media include any sort of tool, usually software, that facilitates group work. Categories of special interest to development projects include:

- project wall (a real wall, not software);
- project website;
- project Wiki;
- modelling tool;
- requirements management tool;
- groupware (software designed specifically to help groups to work).

We should say at once that none of these, not even the next-to-last, is designed explicitly for discovering requirements.

Other tools that can be called groupware include software for negotiation, voting, and issue tracking, as well as communications tools, including anything from email and instant messaging through to videoconferencing.

Let us consider the advantages and disadvantages of each in turn.

12.4.1 Project Wall

Teams traditionally work in a single physical area, for the good reason that this is effective: it facilitates communication and makes management easier.

A simple innovation suggested by the 'agile' school of software development is to provide a conveniently located wall, such as a glass partition, on which everyone in the project can post shared information.

For

At one extreme, a wall can simply display 'top-down' project notices. But a wall encourages a more democratic approach. People can share work-in-progress, such as a large sketch of a current software architecture, or lists of risks, assumptions, issues to be resolved, questions that someone else might be able to answer, and so on.

When a team works in a single area, a project wall can be an excellent choice of groupware. It is on show all the time; it is informal, inviting contributions; it is extremely cheap; and it requires no special training course.

The choice of wall is important. It needs to be in a place where people regularly go – near the office kitchen is often good – and it must be large enough to invite contributions.

The wall can form a good backdrop for team meetings. For instance, a team in one automotive company has an effective weekly stand-up meeting in front of current defect data charts to discuss progress.

Against

A wall is purely passive; its use for discovery depends heavily on the enthusiasm of team members. A wall is not good at displaying large documents or for permanent records (e.g. of published versions of requirements).

People can probably only be expected to pay attention to a few ‘hot’ issues at a time, and responses may arrive over a period of days.

A wall can’t provide the focus and energy of, say, a workshop.

Remote teams can’t see the wall.

Walls can be used well or badly, seriously or frivolously. They aren’t guaranteed to work on every project.

12.4.2 Project Website

Many projects now create their own intranet website (Figure 12.1).

For

A project website is invaluable for sharing standard procedures, dates of meetings and deadlines, document templates, management decisions, training materials and so on.

It is also (unlike a project wall) an ideal place for publishing baselined versions of project documents, including requirements, for all to refer to.

Against

Despite all the advances in website editing, a website can still feel like a ‘read-only’ noticeboard. In practice, only a few people on a project tend to update the website, or perhaps one person is given the role of collating and publishing project information. This makes a project website rather a passive medium and, as such, not very effective at stimulating the discovery of requirements.

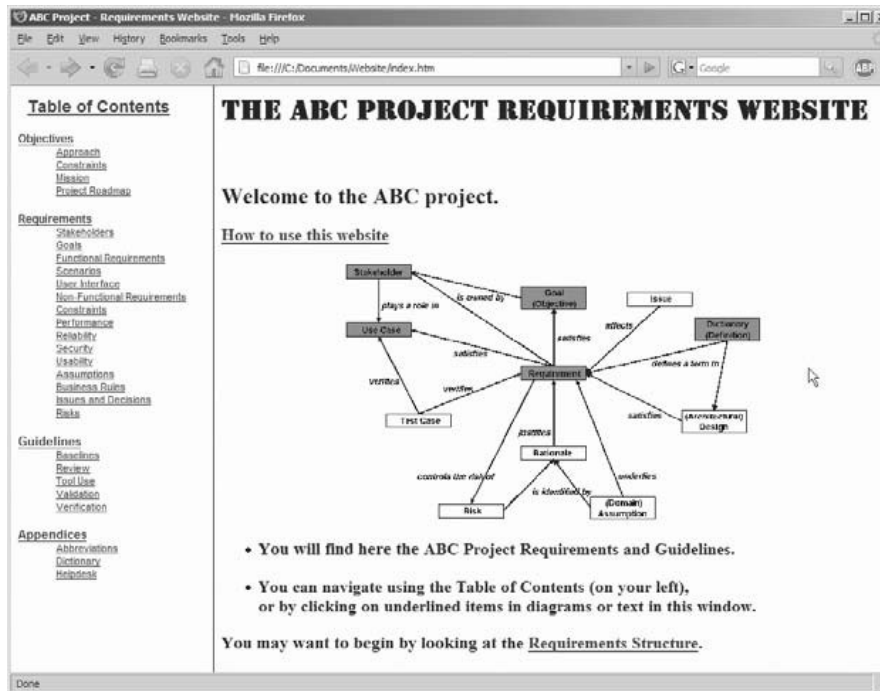


Figure 12.1: A project website.

12.4.3 Project Wiki

A Wiki² is a tool available on a network via an ordinary web browser. Its user interface is a cross between a hypertext and a text editor. Wikis also often keep a history of changes, so people can see who created what, and can revert to an earlier text if need be.

For

A Wiki allows a team to create a shared structure of linked pages, which could describe a project's requirements, design, issues, day-to-day decisions, and so on.

Wikis seem to work best when there are 'topic owners' who actively keep charge of discussion on specific topics.

Some Wikis allow group members to 'watch' for changes to an item, such as the usability requirements page. Anyone may change the page (you may wonder if that is appropriate, if the page depends on specialised knowledge)

²Wiki means 'quick' in Hawaiian. Wikis were invented by Ward Cunningham in 1994. Perhaps the largest Wiki, and certainly one of the best known is Wikipedia (http://en.wikipedia.org/wiki/Main_Page), an encyclopedia edited by the web-using public. It currently contains over 9 million articles written in over 250 languages.

but the usability team are then immediately notified of the change by email. This encourages debate, which may be useful for difficult requirements.

Against

The ease of editing a Wiki gets over some of the weaknesses of the project website as groupware, but also creates challenges of its own:

Wikis sometimes stir up controversy, rather than facilitating agreement.

Wiki pages can be of variable quality, and may quickly be forgotten.

The very ease of editing can encourage team members to keep on 'improving' the requirements, causing scope creep (see Chapter 4). This is uncontrolled 'discovery'.

If not managed by someone, a Wiki's lack of framework can lead to chaos.

The structure of a hypertext is, in a way, very suitable for requirements, as you can create a hierarchy (e.g. qualities include usability and dependability; dependability includes safety, security, and availability, etc) with cross-references for dependencies, and so on. However, this can make it hard to extract a simple, printable document (e.g. a list of all the requirements).

There may be little support for basic project essentials like making a read-only (baseline) copy of the requirements, or recording attributes of requirements (like status and priority).

Current Wikis, therefore, are promising as requirements discovery groupware today, but they need to be used with some care. Perhaps advances in Wiki tools³ will make them more widely suitable for projects.

12.4.4 Modelling Tool

Specific types of modelling tool are described in Appendix C.

The larger modelling tools provide varying degrees of support for group working, starting with the basic ability to share models: the ability to lock individual diagrams, and to notify other users who are currently working on a model. Modelling tools may also offer more Wiki-like facilities, such as history and discussion; or more database-like facilities, such as versioning and linking to requirements.

³For a discussion of the advantages and limitations of Wikis, see: O'Leary, D.E. (2008) Wikis: 'From Each According To His Knowledge', *Computer* 41 (2) 34-41

For

The special thing about a modelling tool as discovery groupware is the model itself, presented to the whole group. Almost any kind of model can stimulate requirements discovery if presented in the right way to the right people. A modelling tool that enables your team to build, discuss, and edit requirements models together should, in theory, be invaluable.

Against

The practice may be rather different. Tools have limitations; they are costly, and take time to set up and to learn. Mainstream tool providers have mostly not given much attention to requirements discovery; there is excellent support for flowcharts and UML class diagrams, but very little for goal and rationale models, for instance. You may have to do awkward things (like putting requirements into comment boxes, or modelling assumptions as classes) to get around tool limitations.

12.4.5 Requirements Management Tool

The use of requirements management (RM) tools for discovery is described in Appendix C. Our question here is how effective they are as discovery *groupware*.

For

As with modelling tools, RM tools offer facilities for sharing requirements, including locking, passwords, history and versioning.

RM tools inherit the benefits of databases – linking information of different kinds, searching, filtering, tracking status, and so on. All of these are valuable in enabling groups to discover requirements, as they facilitate checking for completeness, correctness and consistency.

Now that RM tools are adding the ability to edit models of various kinds, including scenario storyboards, for instance, they are becoming better at supporting discovery directly (and they are turning from ‘pure’ RM tools into project information and project life cycle management tools).

Against

RM tools are primarily designed to manage rather than to discover; and to manage text rather than more direct forms of communication.

Like modelling tools, they can be costly and time consuming.

RM tools can feel unwelcoming to people who do not like software and computers. There can be an air of heavily imposed 'process', offering team members little stimulation to think creatively.

12.4.6 Groupware and Working at a Distance

*There is a certain magic that happens when full-time,
dedicated project members occupy the same space.*

Tom DeMarco *et al* [4]

Having to work at a distance, sometimes across multiple barriers – organisational, linguistic, and cultural, not to mention time zones – is a major challenge for development projects. Distance creates many risks for projects, including a commonly experienced mixture of errors, delays, misunderstandings and rework (Table 12.2).

Numerous 'groupware' tools designed to help with working at a distance have been marketed, using many different technologies including video/web conferencing and telephone conferencing.

Well known examples of groupware include IBM Lotus Sametime, Cisco WebEx, Pixion Picture Talk and Microsoft Live Meeting, among many others.

Such tools offer alternatives to costly and inconvenient travel for actual face-to-face meetings. Some of the tools are themselves costly, though advances in telecommunications and especially the Internet have brought cheaper options.

Table 12.2: Risks of splitting development teams.

Type of barrier	Risks to project	Mitigations
Organisational	Incompatible procedures, documentation styles, notations, resource management. Hence, context not understood, requirements misinterpreted, key staff not available.	Discuss and share procedures and templates; attend training courses together; have some team members co-located; have team leaders visit other team(s) regularly; share a meal, sport, cultural activities; explain requirements and issues visually, not just in text.
National culture	Context not understood, requirements misinterpreted. Some cultures avoid saying anything that might cause someone to lose face.	
Distance, time zone	Meetings become rare/awkward; issues get aired only after long delays; problems build up.	

Collaboration at a Distance



From foggy London to the sunshine of India

In the last few years, organisations have tried to save money on software development by outsourcing. For example, banks have written software specifications in Europe, for coding by the booming software industry in India.

Last year, an English retail company decided to follow suit, passing its requirements to an Indian software house. The software team, experienced in financial software such as banking and insurance, worked hard to meet the tight deadlines, but had difficulty understanding the requirements in the new domain. There was no one to discuss them with. Work was delayed by other unavoidable local factors too, including the monsoon rains.

Meanwhile, the executives of the English company were growing impatient. Why weren't people working faster? They were unaware of the local issues, and could not see how hard the software team was working. Having the 'bandwidth' of communication reduced to written specifications, emails and a few phone calls was seriously limiting progress.

The two organisations realised they both needed to discover which aspects of their different cultures and assumptions the other was having difficulty with. This need was met by having members of the team travel to the other country, taking time to experience the issues. The project never became as straightforward as a local one, but the effort did help to communicate the real requirements. Perhaps paper alone is never enough.

These tools may have started life as attempts to substitute for unmediated group work, which is in some ways the ideal. However, groupware tools now offer useful capabilities that can be better than a traditional face-to-face meeting for some purposes.

Capabilities of groupware include:

- instant messaging ('chat' by typing short messages);
- point-to-point video conferencing ('talking heads' of other participants – two-way or multi-way);
- multi-way voice calls;
- slide shows with voiceover (used in 'webinars');
- virtual whiteboard (shared editing of diagrams and sketches);
- shared desktop (you show participants spreadsheets, etc, while you talk about what you are showing);
- distribution of files (as if you were giving out handouts at a meeting).

New features are constantly being added to groupware tools.

- Some features of groupware tools, such as video conferencing, simply help distributed groups to work almost as well as if they were in one place together. These features do not contribute specifically to discovery: they simply reduce the negative impact of fragmentation on projects. Discovery, however, is a task that is seriously disrupted by distance, as it critically depends on close and continued collaboration. The other major role of requirements, communication of needs, is also very vulnerable to distance (see the box, 'Collaboration at a Distance').
- Other features actively bring people's ideas and contributions together, for example, negotiation and voting (e.g. with Barry Boehm's WinWin approach [5]) and the virtual whiteboard. These can facilitate discovery, and consequently are also useful for groups working on a single site.

Relying on groupware to overcome these obstacles must be seen as a risky approach. You should understand, for instance, that web meeting or video-conferencing facilities will not in themselves overcome the barriers listed in Table 12.2.

12.4.7 The Role of Group Media

Group media, including groupware, have a definite role on the typical project (see Table 12.3, which summarises the discussion in this chapter). Whether their role on your project is essentially as a notice board, or whether you should go further and try to stimulate and manage discovery with a tool, is a decision for you.

Table 12.3: Summary of pros and cons of groupware approaches.

	Groupwork approach	+	–	Points to note
Face-to-Face	Workshop	Power of dialogue and human contact to solve problems, reach agreement	Cost (many salaries, time to plan, need for travel); can be overused	Workshops can create a unique, high-energy environment able to overcome obstacles
Mediated group work	Wall	Shared: democratic, low-cost, informal	Can be ignored; no good for large documents; best if whole team is in one office daily; can be misused	Not suitable for teams split across different sites, or even different buildings on one site
	Website	Available on every PC; ideal for sharing procedures, templates, decisions, baselined documents, etc	Often has read-only feeling, so can be too passive for discovering requirements	Depends on culture; not everyone likes to use software tools as communications medium
	Wiki	Democratic; encourages fine-tuning of text; keeps history	May open debate but fail to close it; can cause endless editing; may make printing etc difficult	
	Modelling tool	Models visible and editable directly; displays concepts and opens them to challenge	Cost and time to set up and learn tool; not many tools help much with requirements	
	RM tool	Many features to help with documenting and managing requirements	Less help for discovery in many RM tools; strong emphasis on text; cost and time to set up and learn	
	Groupware	Attractive to use technology to try to cope with working at a distance	Groupware can simulate proximity, but research shows it is not as good as the real thing	Video conferencing may not overcome barriers of culture and distance

Crossing Cultural Barriers

Engineers in a German company were collaborating on a project with colleagues in China. In Europe, they used an electronic ticketing systems to document and track software bugs.

The Chinese colleagues had real problems with such a system, as using it would mean blaming a colleague, so 'face' would be lost. No amount of persuasion from the European side had any effect.

In the end, the Chinese engineers agreed to assign tickets to European colleagues (who were not part of the project). The Europeans then reassigned the tickets to Chinese co-workers. Getting a ticket assigned from Europe was acceptable, as the strict unwritten rules of 'face' are not assumed to apply in that case.

Group media are not a substitute for face-to-face meetings and for a team simply working closely together.

Some people really don't like having to use software tools, either. Choose tools and techniques with your team. For instance, if no one in the team likes creating Wikis, choose a different approach that they are happy with.

As software evolves, and as teams more often find they have to work at a distance, groupware will probably become increasingly attractive.

12.5 The Bare Minimum from Groups

- Agree any scenarios and issues that affect several stakeholders.
- Make sure the 'bandwidth' of human communication is enough to enable issues to be shared and resolved promptly. That 'bandwidth' is not the same as the promised capabilities of groupware tools.

12.6 Next Steps

Prioritise the requirements, if you have not already done so (Chapter 10).

12.7 Exercise

You are on a multinational project based in California, USA; Sydney, Australia; Bangalore, India; and Rotterdam, The Netherlands. The project is six

months into a three-year development, and is already three months behind schedule. Communication between the teams is proving difficult. The team in Sydney is becoming unhappy with teleconferences at inconvenient times of day, and they feel they are not being listened to. The project manager in California feels that the other three teams are not pulling their weight. The technical leads in Bangalore and Rotterdam are trying to get California to agree the main interfaces they have proposed between them, so they can start on design proper, but feel that California is insisting on making all the decisions.

The project manager has asked you to fix the communications problem between the teams.

- a. What kinds of groupware (software or otherwise) would you advise the project to use, and why?
- b. What other measures would you recommend to improve communications within the project?

12.8 Further Reading

12.8.1 Workshops

1. Gottesdiener, E. (2002) *Requirements by Collaboration: Workshops for Defining Needs*, Boston: Addison-Wesley
Gottesdiener's book is a rich source of proven techniques for workshops. It is accompanied by a website which provides free assets for facilitating workshops: <http://ebgconsulting.com/facassets.php>
2. Beyer, H. and Holtzblatt, K. (1998) *Contextual Design, Defining Customer-Centered Systems*, London: Morgan Kaufmann
Beyer and Holtzblatt's book offers many practical insights.

12.8.2 Working in Groups

3. Cockburn, A. (2006) *Agile Software Development*, 2nd Edition, Boston: Addison-Wesley.
Cockburn's is a useful book, full of practical suggestions. It is worth looking at for ideas on group work even if your project is neither 'agile' nor for software.

4. DeMarco, T., Hruschka, P., Lister, T., McMenamin, S., Robertson, J. and Robertson, S. (2008) *Adrenaline Junkies and Template Zombies: Understanding Patterns of Project Behavior*, New York: Dorset House.
Adrenaline Junkies is for background reading, but you may find it illuminating when trying to understand what games people are playing on projects.
5. Boehm, B. et al, *WinWin Spiral Model and Groupware Support System*, <http://sunset.usc.edu/research/WINWIN/index.html>
Tools change too rapidly to track here, but Barry Boehm's work on the WinWin methodology, which was supported by software tools, was groundbreaking, and the techniques should remain relevant.